Technical Data Sheet

PRODUCT SPECIFICATION AND TYPICAL VALUES

GSB03 | Sulfide solid electrolyte powder | Supplier model: XD-GSB03

Source version: V02 | Source release: March 2026

Typical values are representative characterization, not lot-specific certified results.

Property	Specification	Typical Value	Unit	Method / instrument
Appearance	Gray-white powder; uniform color; no caking	Gray-white powder; uniform color; no caking	-	Visual inspection
D10 particle size	≥0.40	0.65	μm	Laser particle-size analyzer (dry method)
D50 particle size	2.00 ± 1.00	2.19	μm	Laser particle-size analyzer (dry method)
D90 particle size	≤8.50	5.67	μm	Laser particle-size analyzer (dry method)
Water	≤500.00	21.70	ppm	Moisture analyzer
Ionic conductivity	≥6.00	10.12	mS/cm	Electrochemical workstation
Electronic conductivity	$\leq 1.00 \times 10^{-8}$	0.99×10^{-9}	S/cm	Electrochemical workstation
Na impurity	≤200	13.26	ppm	ICP
K impurity	≤200	2.85	ppm	ICP
Ca impurity	≤200	3.53	ppm	ICP
Ni impurity	≤50	ND	ppm	ICP
Cu impurity	≤50	ND	ppm	ICP
Cr impurity	≤50	2.57	ppm	ICP
Zn impurity	≤50	1.25	ppm	ICP
Fe impurity	≤200	2.34	ppm	ICP
Phase identification	Main phase: reference 34-0688	Not reported	-	XRD

TECHNICAL QUALIFICATIONS

Conductivity measurement conditions (temperature, pressure, pellet density and cell configuration) are unspecified. Phase reference 34-0688 is unidentified; XRD phase identification does not establish quantitative phase purity. Ni and Cu are reported as ND; detection limits are unspecified.

SEALED STORAGE AND SHELF LIFE

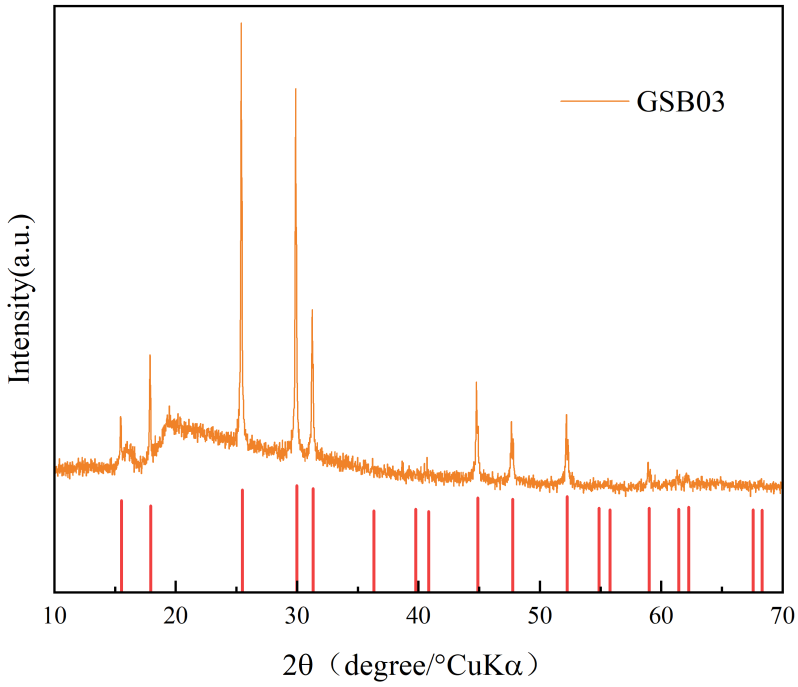
Store sealed and unopened at ≤30% RH and ≤30°C. Shelf life: three months under these conditions. Avoid moisture, heat, strong acids, strong bases and oxidizing agents.

PACKAGING AND OPEN-POWDER HANDLING

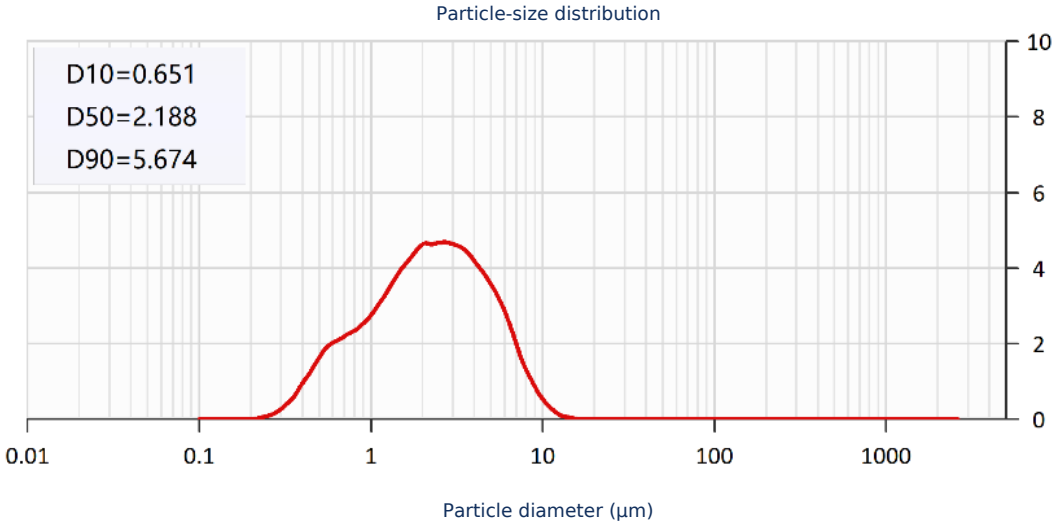
Sealed bottle in a vacuum-packed aluminum-plastic laminate bag; customizable; package sizes start at 10 g per bag. Open powder requires dry inert handling, appropriate protective equipment and SDS guidance. Sealed storage conditions do not define open-powder handling. Dispose of waste according to local regulations.

Characterization Data

GSB03 | X-RAY DIFFRACTION



PARTICLE-SIZE DISTRIBUTION



Typical particle-size table

Diameter (μm)	Content (%)
0.000	0.00
20.00	100.00
30.00	100.00
50.00	100.00
125.0	100.00
200.0	100.00
300.0	100.00
400.0	100.00
500.0	100.00
2600	100.00

REPRESENTATIVE CHARACTERIZATION

The phase reference 34-0688 is unidentified. These representative plots do not establish quantitative phase purity.